Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2024 (NEP-2024)

Subject : Engineering Mathematics **Subject Code : BSC 101**

Time Allowed : 3 Hours **Full Marks : 70**

Answer in your own words.
Answer any five *questions.* Question No. 1 *is compulsory.*
Marks are given in the right margin.

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1. Choose the correct answer in the following: 2×7=14

(i) If A is a 2×2 matrix such that $|A| = 5$ and $|A| \neq 0$ then the value of $|4A|$ is

(a) 20 (b) 25

(c) 80 (d) None of these

(ii) If A and B are invertible square matrices of the same order then $(AB)^{-1}$ =?

(a) AB^{-1} (b) $A^{-1}B^{-1}$

(c) $B^{-1}A$ (d) $B^{-1}A^{-1}$

(iii) Find the slope of a line whose inclination is 60°.

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$

(c) 1 (d) None of these

(iv) The equation of the line that makes intercepts at 2 and −3 on the x-axis and y-axis respectively is represented as

(a) $2x - 3y = 6$ (b) $x - 2y = 3$

(c) $3x - 2y = 6$ (d) None of these

(v) If $\sin x = \frac{1}{6}$ then $\sin 3x$ can be expressed as

(a) $\frac{1}{2}$ (b) $\frac{13}{27}$

(c) $\frac{12}{27}$ (d) None of these

(vi) The first order derivative of $\log_3 x$ is

(a) $\log 3$ (b) $\frac{1}{x}$

(c) $\frac{1}{x(\log 3)}$ (d) None of these

(vii) Find the value of the integral $\int \frac{\sin 2x}{\sin x}\, dx$.

(a) $2\sin x + c$ (b) $2\cos x + c$

(c) $\frac{1}{2}\sin x + c$ (d) $\frac{1}{2}\cos x + c$

2. (a) Prove that $\begin{vmatrix} 1 & b+c & b^2+c^2 \\ 1 & c+a & c^2+a^2 \\ 1 & a+b & a^2+b^2 \end{vmatrix} = (a-b)(b-c)(c-a)$.

(b) Solve the system of equations $x + y + z = 6; 2x + 3y - z = 5; 6x - 2y - 3z = -7$ using Cramer's rule. 7+7

3. (a) Find the equation of the line passing through the point (–2, –4) and perpendicular to the line $3x - y + 5 = 0$.

(b) Reduce the equation $\sqrt{3}x + y + 2 = 0$ to intercept form and find the intercepts on the axes. 7+7

4. (a) Find the values of all trigonometric functions of 120°.

(b) Prove that: $\cos\alpha + \cos\beta + \cos\gamma + \cos(\alpha+\beta+\gamma) = 4\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\beta+\gamma}{2}\right)\cos\left(\frac{\gamma+\delta}{2}\right)$. 7+7

5. (a) Find the second order derivative of $e^{2x}\cos 3x + x^4$.

(b) Obtain the local maxima or local minima of $f(x) = x^3 - 6x^2 + 9x + 15$. Also find the local maximum or local minimum values of $f(x)$. 7+7

6. (a) Evaluate: $\int_{-a}^{a} \sqrt{\frac{a-x}{a+x}}\, dx$.

(b) Calculate the area bounded by the parabola $y^2 = 4ax$ and its latus rectum. 7+7

7. Write short notes on *any four*: 3.5x4=14

(a) Inverse of a matrix

(b) Collinear points

(c) ASTC diagram

(d) Stationary points

(e) Integration by parts

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Diploma 1st Semester Examination, 2024 (NEP-2024)

Subject : Mining Geology I **Subject Code : MIN101**

Time Allowed : 3 Hours **Full Marks : 70**

Pass Marks : 21

Answer in your own words.
Answer any five *questions in which* Question No. 1 *is compulsory.*
The figures in the margin indicate full marks.
All questions carry equal marks.

1. Choose the correct answer: 2×7=14

(i) Gneiss is a
(a) igneous rock (b) sedimentary rock
(c) metamorphic rock (d) None of these

(ii) Which have most hardness according to Mohr's scale?
(a) Diamond (b) Calcite
(c) Feldspar (d) Quartz

(iii) Which sedimentary rock type reacts with dilute hydrochloric acid?
(a) Sandstone (b) Limestone
(c) Shale (d) Breccia

(iv) The uppermost part of the mantle and crust is called
(a) Hydrosphere (b) Atmosphere
(c) Lithosphere (d) Exosphere

(v) Which among the following is an extrusive igneous rock?
(a) Granite (b) Diorite
(c) Tuff (d) Pegmatite

(vi) The term 'aeolian' refers to transport and deposition by
(a) the wind (b) streams
(c) ocean waves (d) pocket gophers

(vii) Which of the following represents the longest time period?
(a) Precambrian (b) Palaeozoic
(c) Mesozoic (d) Permian

2. (i) What are folds? Describe the different types of folds with neat sketches.

(ii) List out and explain different branches of geology. 7+7

3. (i) What do you mean by rock? Discuss different types of rocks with suitable examples of each rock.

(ii) Explain the internal constituents of Earth. 7+7

4. (i) What is the principle of stratigraphy? Describe the physiological subdivisions of India.

(ii) Give the brief classification of igneous rocks based on depth of formation. 7+7

5. (i) What is metamorphism? Discuss the different kinds of metamorphism and provide examples of each.

(ii) Describe the difference between foliated and non-foliated metamorphic rocks, providing examples from India for each category. 7+7

6. (i) What is rock forming mineral? Write down the physical properties of rock forming mineral 'Orthoclase'.

(ii) Explain the role of chemically active fluids in metamorphism and how they lead to the formation of new minerals. 7+7

7. Write short notes on *any four* of the following: 3·5×4=14

(i) Fault

(ii) Erosion

(iii) Poikilitic texture

(iv) Formation of sandstone

(v) Mesozone

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Diploma 1st Semester Examination, 2024 (NEP-2024)

Subject : Computer Aided Engineering Drawing **Subject Code : MIN103**

Time Allowed : 3 Hours **Full Marks : 70**

Pass Marks : 21

Answer in your own words.
Answer any five *questions in which* Question No. 1 *is compulsory.*
The figures in the margin indicate full marks.
All questions carry equal marks.

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1. Choose the correct option / answer the following: 2×7=14

(i) To draw smooth curves of any nature, the drafting instrument used is a/an
(a) mini drafter (b) french curve
(c) template (d) eraser shield

(ii) Parallel lines can be drawn with the help of
(a) mini drafter (b) T-square
(c) pair of set-squares (d) All of these

(iii) The length of scale with R.F. 1/40 to measure up to 6 m will be
(a) 10 cm (b) 12 cm
(c) 15 cm (d) 20 cm

(iv) An area of 36 sq. km is represented by 144 square centimetres on a map. The R.F. is
(a) 1/4 (b) 1/2
(c) 1/5000 (d) 1/50000

(v) If both the front and the top views of a point lie on the same side of the reference line, the point is situated in which of the following angles?
(a) first or second (b) first or third
(c) second or fourth (d) third or fourth

(vi) A solid having minimum number of faces is
(a) tetrahedron (b) triangular prism
(c) square pyramid (d) cube

(vii) The term used by most CAD systems for 'rounding corners' is
(a) chamfer (b) curve
(c) fillet (d) smooth

2. If 1 cm long line on a map represents a real length of 4 m, calculate the R.F. and draw a diagonal scale long enough to measure up to 50 metres. Show a distance of 44.5 m on it. 14

3. An 80 mm long line AB is inclined at 30° to the H.P. and 45° to the V.P. The end A is 20 mm above the H.P. and lying in the V.P. Draw the projections of the line. 14

4. A circular plane of diameter 50 mm is resting on a point of the circumference on the H.P. The plane is inclined at 30° to the H.P. and its centre is 35 mm in front of the V.P. Draw its projections. 14

5. A pentagonal prism of base edge 30 mm and axis 60 mm rests on an edge of its base in the H.P. Its axis is parallel to V.P. and inclined at 45° to the H.P. Draw its projections. 14

6. The front and top views of an angle plate are shown in Fig. A. Draw its isometric view. 14

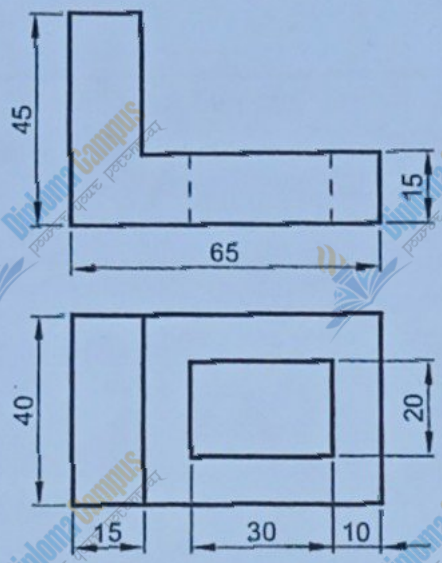


Fig. (A)

7. Pictorial view of an object is shown in Fig. (B). Using first angle projection, draw its (a) front view, (b) top view and (c) right-hand side view. 14

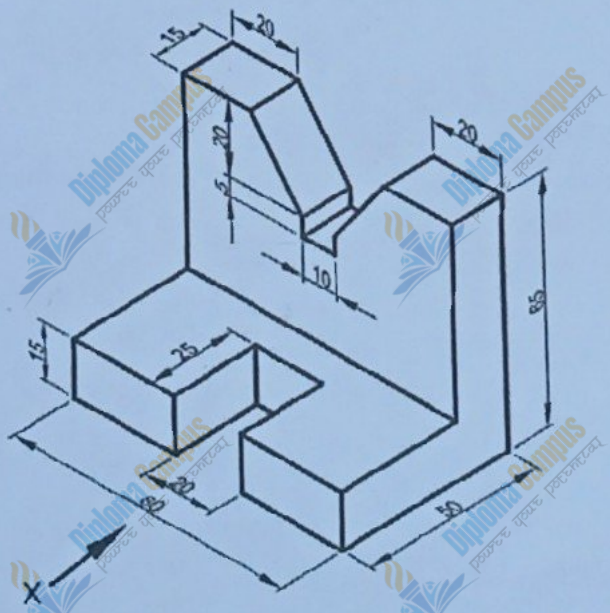


Fig (B)

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Diploma 1st Semester Examination, 2024 (NEP-2024)

Subject : Communication Skills **Subject Code : BSC104**

Time Allowed : 3 Hours **Full Marks : 70**

Pass Marks : 21

Answer in your own words.
Answer any five *questions in which* Question No. 1 *is compulsory.*
The figures in the margin indicate full marks.
All questions carry equal marks.

1. Choose the correct option / answer the following: 2×7=14

(i) How many vowels are there in the English alphabet?
(a) 4 (b) 5
(c) 9 (d) 7

(ii) List out the consonants in the word: Education
(a) d, c, u, n (b) d, c, t, n
(c) d, c, t (d) c, t, n, u

(iii) What is the correct plural form of 'child'?
(a) Childs (b) Childes
(c) Children (d) Childrens

(iv) Choose the correct conjunction: "I like both tea _____ coffee."
(a) but (b) or
(c) and (d) because

(v) What type of sentence is this? "Anna, answer the door"?
(a) Declarative (b) Imperative
(c) Interrogative (d) Exclamatory

(vi) Which sentence is in the passive voice?
(a) The cat chased the mouse. (b) The mouse was chased by the cat.
(c) Chasing the mouse, the cat ran. (d) Mouse is chased by cat.

(vii) Communication saves time in
(a) internal communication. (b) interview.
(c) oral communication. (d) schedule.

2. (a) Define Communication. Discuss various characteristics of successful communication.

   (b) Define Noise. Discuss different types of barriers to communication. 7+7

3. (a) Define Verbal communication. Explain the advantages and disadvantages of written communication.

   (b) What is effective listening? Differentiate between effective and ineffective listening. 7+7

4. (a) What are Non-verbal aspects of communicating? Distinguish between verbal and non-verbal communication.

   (b) What do you mean by Resume? Draft the format of resume. 7+7

5. (a) Read the passage carefully and answer the questions:

   The art of academic writing is not easy to master. It is a formal skill, which requires precision and accuracy and is perfected by continuous and dedicated practice. Academic writing is the skilful exposition and explanation of an argument, which the writer has carefully researched and developed over a sustained period of time. It is a time-consuming activity and demands patience and perseverance. But the joy of reading and sharing with others, one's succinctly composed piece of argument, is incomparable. Before beginning to write, the writer must ask himself a few questions. — Why am I writing? What is it that I intend to share with others? What purpose will my writing serve? Have I read enough about the topic or theme about which I am going to write? If one is hesitant to answer even one of the aforementioned questions, one better not write at all! Because academic writing is a serious activity— it makes one part of a shared community of readers and writers who wish to disseminate and learn from well-argued pieces of writing. The structure of an argumentative essay should take the form of — Introduction (which should be around ten percent of the entire essay), Body (it should constitute eighty percent of the piece) and the Conclusion (again, ten percent of the essay). The introduction should function as the hook which draws the reader in and holds his attention, the body should include cogent and coherently linked paragraphs and the conclusion should re-state the argument and offer a substantial ending to the piece.

   Questions:

   (i) What is academic writing?

   (ii) Why is reading an important part of writing?

   (iii) Why should one ask oneself the questions mentioned in the second passage?

   (iv) What are the components of the structure of an argumentative essay?

   (b) What are different types of reading skills and strategies? 7+7

6. (a) What are the features of fax and email that make them effective tools of speedy communication?

   (b) What are the essential elements of a presentation? 7+7

7. Write short notes on *any four* of the following: 3·5×4=14

(a) Physical barrier

(b) Feedback

(c) Scanning reading skill

(d) Online meeting

(e) Interpersonal communication